

BTEC Level 3 Applied Science • Unit 1 Biology • Answer sheet

Microscopy — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.

A–E

1. $M = \text{image size} / \text{actual size}$. Resolution: smallest distance at which two points can still be seen as separate.

2. $2500\ \mu\text{m}$; $2.5 \times 10^6\ \text{nm}$.

Equation	$1\ \text{mm} = 1000\ \mu\text{m} = 1\ 000\ 000\ \text{nm}$
Substitution	$2.5\ \text{mm} \times 1000$
Working	$2500\ \mu\text{m}$; $2.5 \times 10^6\ \text{nm}$
Answer + unit	$2500\ \mu\text{m}$ and $2.5 \times 10^6\ \text{nm}$

Marking guidance: 1 mark each conversion.

3. Eyepiece magnification $\times$ objective magnification.

4. $18\ \text{mm} = 18\ 000\ \mu\text{m}$; actual = $18\ 000/600 = 30\ \mu\text{m}$.

Equation	$A = I / M$
Substitution	$I = 18\ \text{mm} = 18\ 000\ \mu\text{m}$; $M = 600$
Working	$A = 18\ 000 / 600$
Answer + unit	$30\ \mu\text{m}$

Marking guidance: 1 mark conversion; 1 mark $30\ \mu\text{m}$.

5. $3.0\ \text{cm} = 30\ 000\ \mu\text{m}$; $M = 30\ 000/7.5 = \times 4000$.

Equation	$M = I / A$
Substitution	$I = 3.0\ \text{cm} = 30\ 000\ \mu\text{m}$; $A = 7.5\ \mu\text{m}$
Working	$M = 30\ 000 / 7.5$
Answer + unit	$\times 4000$

Marking guidance: 1 mark conversion; 1 mark $\times 4000$.

6. $40\ \text{mm} = 40\ 000\ \mu\text{m}$; $M = 40\ 000/10 = \times 4000$.

Equation	$M = I / A$
Substitution	$I = 40\ \text{mm} = 40\ 000\ \mu\text{m}$; $A = 10\ \mu\text{m}$
Working	$M = 40\ 000 / 10$
Answer + unit	$\times 4000$

Marking guidance: 1 mark conversion; 1 mark $\times 4000$.

7. 35 mm = 35 000 μm ; actual = 70 μm . [2]

Equation	$A = I / M$
Substitution	$I = 35 \text{ mm} = 35\,000 \mu\text{m}$; $M = 500$
Working	$A = 35\,000 / 500$
Answer + unit	70 μm

Marking guidance: 1 mark conversion; 1 mark 70 μm .

8. Light about 0.2 μm ; TEM much smaller (about 0.1 nm) because electrons have a shorter wavelength. [4]

9. Living specimens / colour / cheaper / easier preparation. [2]

10. The granules are closer together than the resolution. Extra magnification enlarges the blur (empty magnification).